Are spatial spectrums uniquely determined by geometry?

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In this paper, an exposition will be provided for the famous question "Can one hear the shape of a drum?" The question was posed in 1966 by Mark Kac[5]. This presentation will outline the question in detail, provide reasons why a question of this caliber is of particular interest, and describe some of the techinques used to provided some futher understanding of the problem.

I. INTRODUCTION

In physics, the phenomena of a vibrating system producing a frequency spectrum is not an uncommon system to consider. The theories of analyzing spectrums has a wide variety of applications in chaos theory, quantum mechanics, optics, and music. Mark Kac was a Polish-American physicist whose article in *The American Mathematical Monthly* is a famous question investigating a deeper connection between a system and it's frequency spectrum. The question itself is quite simple: Suppose there exists a drumhead constructed in the shape of some region Ω. Can the shape of the drumhead be deduced from knowing the frequency spectrum that would be generated from stricking it?

A. Connecting Geometry and Frequencies

To show the connection between the geometry of a system and its frequency spectrum, consider a string in one dimension[3]. It can be shown (derivations in Fetter) that a wave on a string after being plucked follows the wave equation

$$\frac{\partial^2 \psi}{\partial x^2} = \frac{\partial^2 \psi}{\partial t^2} \tag{1}$$

where $\psi(x,t)$ is the vertical displacement of the string at position x and time t and the speed of the wave on the string is taken to equal to 1. It is also known that the string must satisfy the boundary conditions $\psi(0,t) = 0$, $\psi(L,t) = 0$ for all time t. To solve this partial differential equation, suppose the form $\psi(x,t) = X(x)T(t)$, substitute this into Equation 1, and combine like variables.

$$\frac{1}{X(x)}\frac{\partial^2 X}{\partial x^2} = \frac{1}{T(t)}\frac{\partial^2 T}{\partial t^2} \tag{2}$$

Each side of the equation depends on one variable implying that the whole equation must be equal to a constant in general.

$$\frac{\partial^2 X}{\partial x^2} = -\omega^2 X(x) \tag{3}$$

$$\frac{\partial^2 T}{\partial t^2} = -\omega^2 T(t) \tag{4}$$

From here, it is clear to see that the solution to both of these ODEs are sinusoidal equations of the form

$$X(x) = A\sin(\omega x) + B\cos(\omega x) \tag{5}$$

$$T(t) = C\sin(\omega t) + D\cos(\omega t) \tag{6}$$

Imploying the boundary conditions on the position equation gives conditions on the constants B and ω. The boundary condition $\psi(0,t) = 0$ implies $B = 0$ and $\psi(L,t) = 0$ implies something more interesting:

$$\omega L = n\pi \quad n \in \mathbb{Z}^+ \tag{7}$$

where $\mathbb{Z}^+$ is the positive integers. The boundary conditions have given a condition on the allowable frequencies that can be generated on a string of length L. A general solution to this equation would involve a superposition of the n harmonic solutions given by the previous derivation at varying strengths, but these harmonics are all uniquely defined by the length of the string being plucked. Therefore, someone listening to the sound of a string being plucked could deduce a unique string length L solely from information about its frequency spectrum.

II. KAC'S FORMAL QUESTION

The string was a great example to understand the reason why the geometry of a string might be connected to its allowable frequency spectrum. As a natural progression, one might consider exanding our system to two dimensions and asking about the relationship between the geometry of a two dimensional surface and its spectrum. This is essence of Kac's question. To be more formal, I'll express the question in the same way that Kac presents in his article[5]. Let Ω_1 and Ω_2 be two planar regions bounded by curves Γ_1 and Γ_2. We can consider a wave equation for both regions given by

$$\nabla^2 \psi(x,y,t) = \frac{\partial^2 \psi(x,y,t)}{\partial t^2} \tag{8}$$

and assume a seperation of variables $\psi(x,y,t) = U(x,y)T(t)$. Consider the two spatial wave equations for the regions Ω_1 and Ω_2

$$\nabla^2 U + \lambda_n U = 0 \qquad U(\Gamma_1) = 0 \tag{9}$$
$$\nabla^2 U + \nu_n U = 0 \qquad U(\Gamma_2) = 0 \tag{10}$$

If we assume that all the ω's and ν's are equal for every harmonic n, then does that mean the regions Ω_1 and Ω_2 equal?

III. SOLUTIONS TO THE QUESTION

A. Weyl's Theorem

Historically, answering Kac's question was proving more difficult than it originally seemed. However, Kac was aware that Herman Weyl proved a theorem closely related to his question[5] (in fact, this proof spurred Kac's question about the drums). Weyl proved the spectrum satisfying the spatial differential equation follows the relationship[6]

$$\lim_{n\to\infty} \frac{\lambda_n}{n} = \frac{4\pi}{|\Omega|} \tag{11}$$

where $|\Omega|$ is the area of the region of interest. Instead of outlining the proof, consider proving this for the example of a rectangular drumhead. The example is outlined in McCorkle[6], but I've expressed the solution in a more personally satisfying presentation.

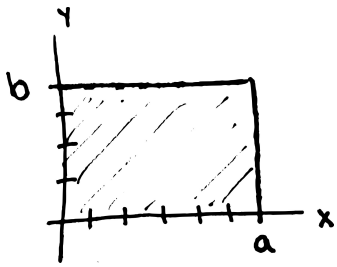

FIG. 1: Rectangular region Ω

Consider Equation 9 for the rectangular region with side lengths a and b with the boundary conditions $U(0,0) = U(a,0) = U(0,b) = U(a,b) = 0$(see Figure 1). Since the spatial equation is two dimensional, employ another seperation of variables to get a single equation in both the x and y directions.

$$\frac{\partial^2 X}{\partial x^2} = -\mu X(x) \qquad X(0) = X(a) = 0 \tag{12}$$
$$\frac{\partial^2 Y}{\partial y^2} = -\gamma Y(y) \qquad Y(0) = Y(b) = 0 \tag{13}$$

where $\lambda_{kl} = \mu_k + \gamma_l$. From here, it is clear to see the solution for each direction is the solution to the original string problem with different variables

$$X(x) = B\sin\left(\frac{k\pi}{a}\right) \qquad \mu_k = \frac{k^2\pi^2}{a^2} \tag{14}$$
$$Y(y) = D\sin\left(\frac{l\pi}{a}\right) \qquad \gamma_l = \frac{l^2\pi^2}{b^2} \tag{15}$$

To be able to relate the integer n in Weyl's Theorem to the pair of (k,l), it's helpful to think in the (k,l) state space. Consider the expression for the eigenvalue λ below

$$\lambda_{kl} = \frac{k^2\pi^2}{a^2} + \frac{l^2\pi^2}{b^2} \tag{16}$$
$$\frac{k^2\pi^2}{\lambda_{kl}a^2} + \frac{l^2\pi^2}{\lambda_{kl}b^2} = 1 \tag{17}$$

where the second equation is written in the form of the ellipse equation. In the (k,l) state space, the above equation can be imagined geometrically as a quarter ellipse where the semi-major and -minor axes are given by

$$\text{semi-major} = \frac{a\lambda_{kl}}{\pi} \tag{18}$$
$$\text{semi-minor} = \frac{b\lambda_{kl}}{\pi} \tag{19}$$

since (k,l) are only positive. We are making the assumption that our eigenvalues are large enough to let (k,l) be considered continuous variables, which is a technique that is often used in statistical mechanics when discussing state spaces. With this approximation, the λ_{kl} can now be identified by λ_n where the integer n corresponds to the "radius" of the ellipse in state space. This allows us to not worry about the degeneracy of the eigenvalues and only focus on the number of distinct eigenvalues.

From here, there's an interesting trick that can be employed thanks to our state space representation. Consider wanting to write an expression for the integer n. You can convince yourself that n can be found from counting all the distinct eigenvalues that appear before λ_n. That's fairly trivial in the case of the 1 dimensional string, but the idea is very useful for our rectangular region. Counting the number of distinct eigenvalues in (k,l) space amounts to counting all the radii below the maximum. Still assuming we're treating (k,l) as continuous variables, then this amounts to taking the area of the ellipse in the state space.

$$A = \frac{1}{4}\pi\left(\frac{a\sqrt{\lambda_n}}{\pi}\right)\left(\frac{b\sqrt{\lambda_n}}{\pi}\right) = \frac{ab\lambda_n}{4\pi} \tag{20}$$

Therefore, the expression for the integer n in the limit that $n \to \infty$ is given by the area of the ellipse in state space.

$$\lim_{n\to\infty} n = \frac{ab\lambda_n}{4\pi} \tag{21}$$

Rearranging the variables leads to the expression

$$\lim_{n\to\infty} \frac{\lambda_n}{n} = \frac{4\pi}{ab} \tag{22}$$

which is in the form of Equation 11 where $|\Omega| = ab$. Since the area of our rectangular region is ab, then Weyl's Theorem has been returned. Therefore, the frequency spectrum for any region is uniquely defined by the magnitude of the region.

B. 2D Isospectral Regions

In 1992, Gordon et. al.[4] constructed the first example of 2-dimensional isospectral regions of different shapes. This was accomplished by using a theorem from Toshikazu Sunada and a proposition by Pierre Berard. Although the rigorous theorem and proposition are difficult for me to understand, I've provided them below for completeness.

Theorem 1 (Sunada) *Let M be a Riemannian manifold upon which a finite group G acts by isometries; let H and K be subgroups of G that act freely. Suppose that H and K are almost conjugate, i.e., there is a bijection $f : H \to K$ carrying every element h of H to an element $f(h)$ of K that is conjugate in G to h. Then the quotient manifolds $M_1 = H\backslash M$ and $M_2 = K\backslash M$ are isospectral.*

Proposition 1 *Let G be a group with subgroups H and K, and suppose that $T : L^2(H\backslash G) \to L^2(K\backslash G)$ is a unitary intertwining operator. Let V be a Hilbert space on which G acts unitarily. Then T induces an isometry $V^H \to V^K$.*

For example, Gordon and Webb mention Robert Brooks[1] and Peter Buser[2] used these groups G to construct the isospectral regions not in 2 dimensions. Both Brook and Buser reference the group $\mathbb{Z}_8^* \ltimes \mathbb{Z}_8$ (the semi direct product of the multiplicative group of integers mod 8 and the additive group of eight elements) with group operation

$$(a, b) \circ (c, d) := (a \cdot c, a \cdot d + b) \bmod(8) \tag{23}$$

This group admits two subgroups that are almost conjugate

$$H := \{(1, 0), (3, 0), (5, 0), (7, 0)\} \tag{24}$$
$$K := \{(1, 0), (3, 4), (5, 4), (7, 0)\} \tag{25}$$

Therefore, the groups can be used to construct isospectral regions which they outlined in their respective papers.

A simpler explanation of Sunada's theorem was provided by Gordon and Webb[3]. The essence of the theorem takes advantage of unique representations of the groups G discussed by Brooks and Buser. In particular, building isospectral regions involved considering linear and permutation represenations. A linear representation of a group takes the elements of a group and represents each element as matrices while permutation representations represent each element of a group as a permutation on elements in a set. Sunada's theorem use the fact that *certain groups can have unique permutation represenations, but the linear representations of the permutations are identical.*

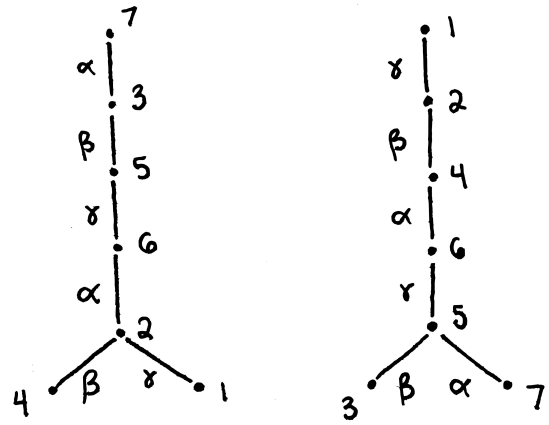

FIG. 2: Cayley graphs of the two permutation represenations of the group. The permutation group includes the elements α, β, γ that act as permutations on the set X of elements 1 through 7.

In Gordon and Webb's case, they discuss a set of unique elements α, β, γ. The unique elements and their products construct a group G with almost conjugate subgroups. The group G presents two unique permutation representations on a set $X = \{1, 2, 3, 4, 5, 6, 7\}$ seen in Figure 2. These permutation representations admit identical linear representation. Therefore, the permutation representations of the group G can be used to construct isospectral regions. They begin with a model triangle whose edges are indentified with the elements α, β, γ and label the triangle with one of the elements in X. The isospectral regions can be generated by reflecting the triangle over the edge labelled α, β, γ using the Cayley graphs in Figure 2 as instructions. The reflected triangle is then labelled the next number in the Cayley table, and the process continues. For example, following the left graph in Figure 2 means the triangle labelled 7 reflects over the edge identified with the element α. The reflected triangle is labelled 3, and the process continues until all the elements in the set X have been permutated. Applying both Cayley tables to the similar model triangles constructs the unique regions in Figure 3 that are isospectral.

To confirm the regions are isospectral, one must show each region exhibit standing wave solutions with identical frequencies. This can be done by considering a particular wave solution on one region and checking if the solution can be cleverly mapped onto the second region. Berard's

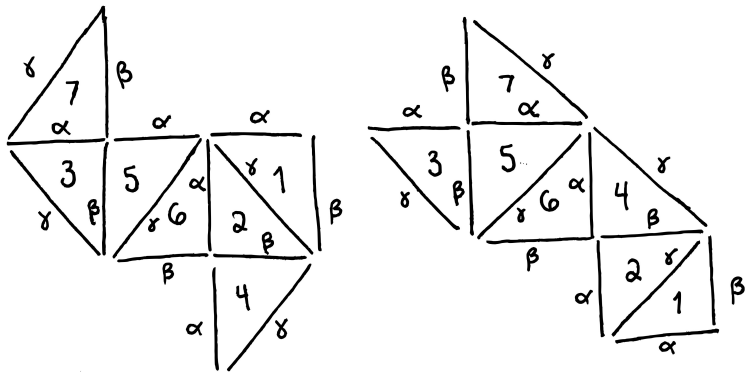

FIG. 3: The two unique isospectral regions generated by following the procedure of Sunada's Theorem. The region on the left is generated by following the left Cayley table in Figure 2 and the region on the right is constructed from the Cayley graph on the right.

proposition allowed Gordon and Webb to map a wave solution from one region to the other. Consider a wave solution on each of the triangles that are continuous between triangles and zero on the boundary. In Figure 4, the triangles in the left region are labelled $A, B, ..., G$ associated with the particular portion of the wave solution in that region (U_A, U_B, etc.). The sides have also been color coded with the unique elements α (red), β (blue), and γ (green).

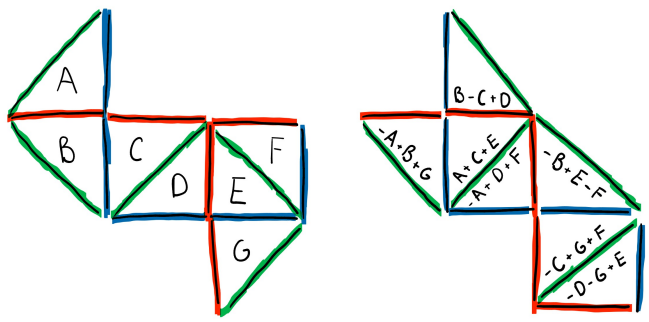

FIG. 4: Mapping a wave solution of a particular frequency from the left to the right region with a clever choice of mapping provided by Gordon and Webb.

In each triangle on the right region, there is a superposition of the solutions $A, B, ...$ To prove this is a wave solution with identical frequency, it can be shown the complicated superposition solution follows continuous mapping properties between the triangles (e.g. $A = B$ on the red boundary) and boundary conditions are zero. For example, consider the two triangles labelled $-C+G+F$ and $-D-G+E$ on the bottom right of the right region. We must show the boundary sides are zero and continuous mapping exist between the two triangles on their shared green side. Let's check if the red side of $-D-G+E$ goes to zero. From the left region, we can see that the red side of G is defined to be zero because it's on the boundary. Also, we can see that D and E share the red side, meaning they're the same. Therefore, $-D-G+E$ goes to zero on the red boundary. The same can be shown for the blue side of $-D-G+E$ and the red side of $-C+G+F$. We also need to show the green side maps nicely between the two triangles. The green side of G is defined to be zero because it's again a boundary. On the left region, C/D and E/F share the green side, so $C = D$ and $E = F$ on the green side. On the right region, $-C+F$ must equal $-D+E$ which is the case using the definitions from the left region. These properties can be shown for every triangle in the right region which means a wave solution with identical frequency exists on both regions.

IV. CONCLUSION

Although Herman Weyl was able to prove the spectrum of a drum is unique defined by the area of the drumhead, Gordon et. al. constructed a counterexample of the two isospectral regions that are not isometric. This is thanks to Sunada's theorem of almost conjugate subgroups and Berard's proposition allowing for transplanting of wave solution between the two regions. However, this counterexample has only seemed to uncovered more interesting questions. For instance, can it be sown in general that particular areas of the region never exhibit isospectral regions? How many isospectral regions can be constructed for a particular geometry? Is it more likely for regions of a certain area to permit isospectral regions or not? These questions will further help physicist to understand the dynamics of systems were the key interest are the spectrum generated.

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